



The Companion Planting Chart

What to Plant Together and What to Keep Apart

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Open-pollinated • Community-grown • Food sovereignty for everyone

The Companion Planting Chart

What to Plant Together and What to Keep Apart

by Seedwarden

Some plants help each other grow. Others fight for the same nutrients, attract the same pests, or release chemicals that stunt their neighbors. This chart tells you which is which.

Print it. Pin it to the wall above your potting bench. Use it every time you plan a bed.

Companion Planting Matrix

Format: **Plant** | Companions (reason) | Bad Neighbors (reason)

FRUITING VEGETABLES

Tomato

- Companions: basil (repels aphids + whiteflies, may improve flavor), carrot (loosens soil around roots), marigold (kills root-knot nematodes), parsley (attracts hoverflies), borage (deters tomato hornworm), nasturtium (trap crop for aphids), garlic (fungal disease suppression), celery, pepper, spinach
- Bad Neighbors: cabbage family (stunts both), fennel (allelopathic — inhibits growth), potato (shares blight), corn (shares tomato fruitworm), kohlrabi, walnut trees

Pepper (Sweet + Hot)

- Companions: basil (repels aphids + spider mites), tomato (similar needs, no conflict), carrot, onion (deters aphids), spinach (living mulch), marigold, oregano, parsley
- Bad Neighbors: fennel (allelopathic), kohlrabi (stunts peppers), beans (may shade out peppers), apricot trees

Eggplant

- Companions: beans (nitrogen fixation), pepper, marigold (nematode control), thyme (repels garden moths), spinach (ground cover), catnip (deters flea beetles), nasturtium (trap crop)
- Bad Neighbors: fennel, walnut trees

Cucumber

- Companions: beans (nitrogen), corn (windbreak + trellis), peas, radish (trap crop for cucumber beetles), sunflower (pollinator attraction + trellis), dill (attracts predatory wasps), marigold, nasturtium, lettuce, borage
- Bad Neighbors: potato (shares blight + competes), sage (inhibits growth), melon (cross-pollination confusion, shared pests), aromatic herbs in excess

Squash (Summer + Winter)

- Companions: corn (Three Sisters — structural support for beans), beans (nitrogen for corn + squash), nasturtium (trap crop for squash bugs), marigold, radish (trap crop), borage (attracts pollinators), oregano, dill
- Bad Neighbors: potato (competes + shared blight risk), other cucurbits nearby (shared vine borers)

Melon (Watermelon + Cantaloupe)

- Companions: corn (wind protection), sunflower (pollinator magnet), nasturtium, marigold, radish (fast harvest, frees space), lettuce (early-season ground cover)
- Bad Neighbors: potato, other cucurbits in tight spacing (shared pests concentrate)

Corn

- Companions: beans (fix nitrogen corn needs), squash (shades soil, deters raccoons), peas, cucumber, melon, sunflower (windbreak on edges), dill, marigold
- Bad Neighbors: tomato (shares earworm/fruitworm), celery (heavy feeder competition)

ROOT VEGETABLES

Carrot

- Companions: onion (masks carrot scent from carrot fly), leek (same — mutual fly deterrence), rosemary (repels carrot fly), sage (repels carrot fly), lettuce (marks rows, loosens surface), tomato, pea, chive, radish (breaks soil crust for carrot seedlings)
- Bad Neighbors: dill (cross-pollinates, may stunt), celery (competes in same root zone), parsnip (shares carrot fly)

Potato

- Companions: beans (nitrogen fixation), corn, cabbage, marigold (nematode suppression), horseradish (at corners — disease resistance), pea, garlic (fungal suppression)
- Bad Neighbors: tomato (shares late blight — #1 risk), cucumber, squash, sunflower (allelopathic to potato), fennel, raspberry

Beet

- Companions: onion, garlic, lettuce (different root depths), cabbage family, bush beans, kohlrabi
- Bad Neighbors: pole beans (mutual growth inhibition), mustard (related — shared pests), charlock

Radish

- Companions: carrot (radish breaks crust, harvested before carrots need space), cucumber (trap crop for cucumber beetles), pea, lettuce, nasturtium, spinach, beans, squash
- Bad Neighbors: hyssop (inhibits radish growth), grapes

Turnip

- Companions: pea (nitrogen), onion, garlic, nasturtium, vetch
- Bad Neighbors: potato (competes in root zone), mustard (same family — shared clubroot), other brassicas in tight rotation

Parsnip

- Companions: radish (marks slow-germinating rows), onion, garlic, pea, bush beans
- Bad Neighbors: carrot (shares carrot fly — don't plant adjacent), celery, dill

Sweet Potato

- Companions: beans (nitrogen), corn, dill, thyme (ground beetle deterrent), marigold
- Bad Neighbors: squash (vine competition for space), tomato (shares soil-borne pathogens)

ALLIUMS

Onion

- Companions: carrot (mutual fly deterrence), beet, lettuce, tomato, pepper, strawberry (deters slugs), chamomile (improves flavor + vigor), cabbage family (deters cabbage worm)
- Bad Neighbors: beans (onion inhibits bean growth), peas (same — alliums stunt legumes), asparagus, sage

Garlic

- Companions: tomato (fungal suppression), pepper, beet, lettuce, carrot, rose (deters aphids), fruit trees (deters borers), spinach, cabbage
- Bad Neighbors: beans (growth inhibition), peas (same), asparagus, strawberry (some sources — conflicting evidence)

LEGUMES

Bush Beans

- Companions: corn (nitrogen exchange), potato, carrot, cucumber, celery, strawberry, squash, beet, cabbage, cauliflower, marigold, rosemary (repels bean beetles), nasturtium (trap crop)
- Bad Neighbors: onion (alliums stunt legumes), garlic, leek, chive, fennel, pepper (may shade)

Pole Beans

- Companions: corn (classic trellis + nitrogen partnership), squash, radish, marigold, nasturtium, rosemary
- Bad Neighbors: beet (mutual inhibition), onion, garlic, sunflower (allelopathic to pole beans), fennel

Pea

- Companions: carrot, radish, turnip, corn, cucumber, bean, spinach (appreciates pea's nitrogen), lettuce, mint (repels pea moth)
- Bad Neighbors: onion (alliums inhibit legumes), garlic, leek, chive, potato (in some soils)

BRASSICAS

Cabbage

- Companions: onion (deters cabbage maggot), garlic, dill (attracts predatory wasps targeting cabbage worm), celery (repels cabbage white butterfly), thyme (repels cabbage worm), nasturtium (trap crop), chamomile (improves growth), mint (deters cabbage moth), beet
- Bad Neighbors: strawberry (stunts both), tomato (stunts both), pole beans, grape

Broccoli

- Companions: onion, garlic, celery, beet, chamomile, dill, rosemary (repels cabbage fly), nasturtium, marigold
- Bad Neighbors: tomato (heavy feeder competition), strawberry, pole beans, lettuce (some evidence of bolting trigger)

Cauliflower

- Companions: celery, bean (bush), onion, garlic, beet, spinach, dill, chamomile
- Bad Neighbors: tomato, strawberry, pole beans, pea in some conditions

Kale

- Companions: beet, celery, onion, garlic, dill, nasturtium, marigold, chamomile, potato (kale deters some potato pests)
- Bad Neighbors: strawberry, tomato, pole beans

LEAFY GREENS

Lettuce

- Companions: carrot (different root levels), radish (fast shade cover), onion, garlic (pest deterrence), strawberry, chive, dill, cilantro, mint (slug deterrent), marigold, tall plants for afternoon shade in summer
- Bad Neighbors: celery (competition in similar root zone), parsley in tight quarters

Spinach

- Companions: strawberry (complementary root depths), pea (nitrogen + shade), beans, brassicas, celery, onion, garlic, radish, nasturtium, tomato (spinach as living mulch)
- Bad Neighbors: potato in tight spacing, fennel

CELERY & ASPARAGUS

Celery

- Companions: tomato, bean (bush), cabbage family, spinach, onion, leek, nasturtium, marigold, dill
- Bad Neighbors: carrot (root zone competition), parsnip (same zone), corn (heavy feeder competition), potato

Asparagus

- Companions: tomato (kills asparagus beetle), parsley (vigorous growth pairing), basil, marigold, nasturtium, dill, comfrey (nutrient accumulator)
 - Bad Neighbors: onion (stunts asparagus), garlic, potato, fennel
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HERBS

Basil

- Companions: tomato (#1 pairing — repels aphids, whiteflies, may improve flavor), pepper, asparagus, oregano, marigold
- Bad Neighbors: sage (conflicting moisture needs), rue, thyme in some conditions

Dill

- Companions: cabbage family (attracts parasitic wasps for cabbageworm), lettuce, onion, cucumber, corn, asparagus
- Bad Neighbors: carrot (cross-pollinates if both flower, stunts carrot), tomato when mature (inhibits tomato growth — young dill is fine)

Cilantro

- Companions: tomato, pepper, spinach, lettuce, beans, pea, anise (bolts slowly near anise)
- Bad Neighbors: fennel (cross-pollination issues), lavender (conflicting water needs)

Parsley

- Companions: tomato, asparagus (classic pairing), pepper, carrot (same family but beneficial), corn, bean, rose
- Bad Neighbors: lettuce in tight quarters, mint (competes aggressively)

Rosemary

- Companions: cabbage family (repels cabbage fly), bean (repels bean beetles), carrot (repels carrot fly), sage, lavender, thyme (Mediterranean guild)
- Bad Neighbors: cucumber (growth inhibition), potato, pumpkin

Thyme

- Companions: cabbage family (repels cabbage worm), eggplant, tomato, strawberry (deters worms), rosemary, lavender
- Bad Neighbors: basil in some conditions (moisture mismatch), chive

Sage

- Companions: cabbage family (repels cabbage moth + carrot fly), rosemary, carrot, strawberry, tomato, bean
- Bad Neighbors: cucumber (inhibits cucumber growth), onion, rue, alliums generally

Oregano

- Companions: tomato, pepper, squash (general pest deterrent), basil, bean, broccoli, cabbage
- Bad Neighbors: mint (competes aggressively if not contained)

Mint *(always container-plant or isolate)*

- Companions: cabbage (repels cabbage moth), tomato (repels aphids), pea (repels pea moth), lettuce (slug deterrent)
- Bad Neighbors: parsley, chamomile (mint will smother), everything if not contained — invasive root system

Chamomile

- Companions: cabbage family (improves growth + flavor), onion (improves flavor), brassicas generally, cucumber, wheat
- Bad Neighbors: mint (smothered by mint), other aggressive spreaders

Lavender

- Companions: rosemary, thyme, sage (Mediterranean guild — same needs), fruit trees (pollinator draw), rose (repels aphids)
 - Bad Neighbors: cilantro, basil (conflicting water needs — lavender wants dry, basil wants moist)
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FLOWERS

Marigold (French + African)

- Companions: tomato (#1 flower companion — kills root-knot nematodes), pepper, eggplant, potato, bean, squash, cucumber, rose — plant as borders everywhere
- Bad Neighbors: beans in some studies (slight inhibition), cabbage (marginal)

Nasturtium

- Companions: tomato, cucumber, squash, cabbage family, bean, radish, fruit trees — supreme trap crop for aphids, whiteflies, squash bugs
- Bad Neighbors: cauliflower in some conditions

Sunflower

- Companions: cucumber (trellis + pollinator magnet), corn (windbreak), squash, melon, lettuce (shade), bean, pea
- Bad Neighbors: potato (allelopathic to potato — confirmed), pole beans (allelopathic)

Zinnia

- Companions: tomato, pepper, squash, melon, cucumber — attracts pollinators + predatory insects (ladybugs, parasitic wasps)
- Bad Neighbors: none significant — zinnia is a universal pollinator draw

Borage

- Companions: tomato (deters tomato hornworm), squash (attracts pollinators), strawberry (improves yield + flavor — classic pairing), cucumber, cabbage
- Bad Neighbors: none significant — borage is nearly universal-positive

Quick Reference: The Classic Combos

Three Sisters (Corn + Pole Beans + Squash)

- Corn provides structure for bean vines
- Beans fix nitrogen for heavy-feeding corn and squash
- Squash shades soil with broad leaves, deters raccoons with prickly stems
- Plant corn first, beans when corn is 6", squash when beans are climbing

Pizza Garden (Tomato + Basil + Oregano + Pepper)

- Basil repels tomato pests, oregano provides general pest deterrence
- All thrive in same conditions: full sun, consistent moisture, warm soil
- Plant in a 4x4 raised bed: tomatoes center, peppers corners, herbs between

Salsa Garden (Tomato + Pepper + Onion + Cilantro)

- Onion deters aphids from tomato and pepper
- Cilantro attracts beneficial insects, bolts to coriander for seed harvest
- Add garlic at edges for fungal suppression

Pest Barrier Trio (Marigold + Nasturtium + Borage)

- Marigold kills nematodes in soil
- Nasturtium draws aphids and whiteflies away from crops (sacrifice plant)
- Borage attracts pollinators and predatory insects
- Border every bed with this combination

Root Depth Guild (Carrot + Onion + Lettuce)

- Carrot: deep taproot
- Onion: shallow-medium bulb
- Lettuce: shallow fibrous roots
- No competition — three crops from one square foot, plus onion repels carrot fly

Mediterranean Herb Block (Rosemary + Thyme + Sage + Lavender + Oregano)

- All want: full sun, lean/dry soil, excellent drainage
- Plant together in a separate bed or raised mound with gravel mulch
- Do NOT mix with moisture-loving herbs (basil, cilantro, parsley, mint)

Quick Reference: Common Mistakes

1. Tomato next to Potato

They're the same family. They share late blight, early blight, and Colorado potato beetle. One infection takes out both crops. Separate by at least 20 feet.

2. Onion/Garlic next to Beans/Peas

Alliums chemically inhibit legume growth. This is one of the most reliable negative pairings. No exceptions.

3. Dill next to Carrots

Same family (Apiaceae). They cross-pollinate, and mature dill releases compounds that stunt carrot roots. Young dill near tomatoes is fine — mature dill is not.

4. Fennel next to Anything

Fennel is allelopathic to nearly every garden vegetable. Grow it in a pot or isolated corner. This is not exaggeration — fennel actively inhibits growth of most plants within root range.

5. Mint Without Containment

Mint's underground runners will colonize an entire bed in one season. Always grow in a buried pot or dedicated container. It is a companion to many plants — from 3 feet away, in a pot.

6. All Brassicas in One Bed

Concentrating cabbage, broccoli, kale, and cauliflower in one spot is a pest buffet. Cabbage worm, cabbage looper, and flea beetles will devastate everything at once. Interplant with onion, dill, and thyme to break the monoculture.

7. Walnut Trees Near Anything

Black walnut produces juglone, toxic to tomatoes, peppers, eggplant, potato, and many others. Toxicity extends to the drip line and beyond. If you have a walnut tree, test your soil.

The Science: Why Companion Planting Works

Allelopathy — Some plants release chemicals from roots, leaves, or decomposing material that inhibit or promote growth of nearby plants. Black walnut (juglone) and fennel are negative examples. Marigold root exudates killing nematodes is a positive example. These effects are real and measurable.

Nitrogen Fixation — Legumes (beans, peas, clover) host Rhizobium bacteria on their roots that convert atmospheric nitrogen into plant-available form. This benefits neighboring and subsequent crops. The effect is strongest when legume roots decompose — don't pull bean plants after harvest, cut them at soil level.

Trap Cropping — Sacrificial plants that attract pests away from your main crop. Nasturtium draws aphids. Radish draws flea beetles away from brassicas. Blue Hubbard squash draws squash vine borers away from zucchini. The trap crop takes the hit so your harvest doesn't.

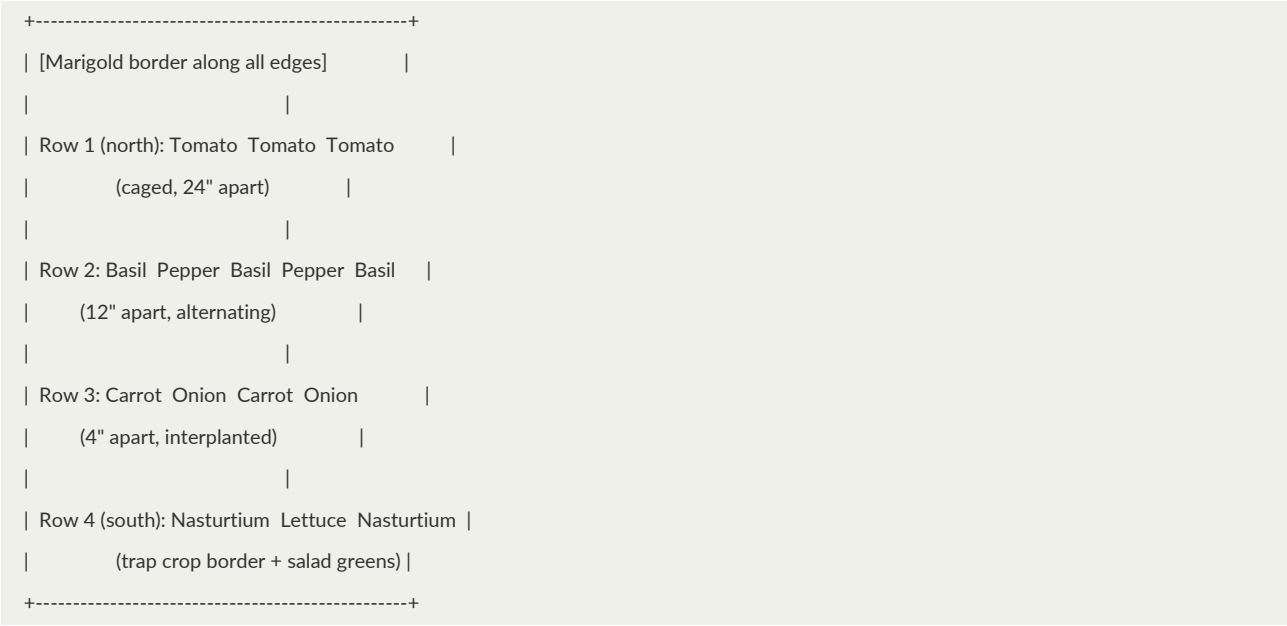
Beneficial Insect Attraction — Flowers and herbs draw pollinators (bees, butterflies) and predatory insects (ladybugs, lacewings, parasitic wasps) that control pest populations. Dill, fennel, yarrow, and members of the carrot family are top attractors for parasitic wasps. Borage and zinnia bring pollinators.

Physical Effects — Tall corn shades lettuce and spinach in summer heat. Squash leaves shade soil, reducing evaporation and weed growth. Dense plantings of thyme or oregano create living mulch. Sunflowers serve as windbreaks and natural trellises.

Scent Masking — Aromatic plants (basil, onion, garlic, rosemary) can mask the scent signature of nearby crops, making it harder for pests to locate their target by smell. This is why interplanting herbs throughout the garden works better than a single herb patch.

Planting Layout Examples

Layout 1: The 4x8 Raised Bed — Summer Producers



Why it works: Tomatoes in back (tallest, north side so they don't shade others). Basil between peppers repels shared pests. Carrot-onion guild uses different root depths and mutual pest deterrence. Nasturtium border catches aphids before they reach crops.

Layout 2: The Three Sisters Mound (10x10 area)

Mound layout (repeat in grid, 5' between mounds):

[Corn x4]

in center, 12" apart

[Pole Bean x4]

planted 6" from corn base

after corn reaches 6" tall

[Squash x2]

planted 18" from mound center

vines radiate outward

Edge: Sunflower sentinels on north side (windbreak)

Border: Marigolds around entire 10x10 area

Why it works: Corn supports beans. Beans feed nitrogen to corn and squash. Squash leaves shade the ground, conserve moisture, and prickly stems deter animal pests. Sunflowers add wind protection and pollinator draw.

Layout 3: The 4x4 Herb + Salad Bed (Cool Season)

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| Dill (1)  Chamomile (1) |
|           |
| Lettuce  Spinach  Lettuce |
| (succession plant every 2wk) |
|           |
| Radish  Carrot  Radish  |
| (radish marks carrot rows) |
|           |
| Pea (along south trellis) |
| Onion sets between pea plants |
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Why it works: Dill and chamomile attract beneficial insects. Lettuce and spinach thrive in pea-nitrogen-enriched soil and appreciate dappled shade from peas. Radish breaks soil crust for slow-germinating carrots and is harvested before carrots need the space. Onions between peas is an exception to the allium-legume rule — spring onion sets are short-season enough to coexist.

Disclaimer

This chart is for educational and planning purposes. Companion planting results vary by region, soil type, microclimate, specific cultivar, and growing conditions. The information here is compiled from university extension research, published agricultural studies, and generations of grower experience. It is not a guarantee of specific outcomes. Test in your own garden, observe what works in your conditions, and adjust accordingly.

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The Soil Building Cheat Sheet — Cover crops, amendments, and no-till methods that actually work. Rebuild your soil in 2 seasons.

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